

El Moustpha Cheikh JIDDOU

Data & AI Engineer – Production Pipelines, RAG/LLM Systems and MLOps

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Nouakchott, Mauritania

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Portfolio

PROFESSIONAL EXPERIENCE

Data & AI Engineer – Richat Partners

August 2026 – Present

Nouakchott, Mauritania

- Design and implementation of AI and data solutions for strategic consulting missions.
- Development of data pipelines and machine learning models to support decision-making in digital transformation, investment, and strategy projects.
- Collaboration with multidisciplinary teams to translate business needs into technical solutions.
- Contribution to the firm's AI offerings: LLM integration, RAG systems, predictive analytics, and data-driven advisory.

Data Systems & IT Architecture Intern – Banque Nationale de Mauritanie (BNM)

February 2025 – Present

Nouakchott, Mauritania

- Designed and implemented a centralized enterprise Data Warehouse backed by robust ELT pipelines, automated flow orchestration, and predictive models.
- Implemented a RAG (Retrieval-Augmented Generation) architecture with PostgreSQL/pgvector to index and query official banking documentation without data leakage.
- Benchmarked several LLMs (GPT-4o, Claude 3.5, Gemini 1.5, local Llama) to balance performance, infrastructure cost, and strict confidentiality.

Assistant Engineer Intern – Mining & Geotechnical Department – Kinross Gold Corporation (Tasiast)

July 2025 – September 2025

Tasiast, Mauritania

- PV short-term forecasting: prepared minute-level data (lags, ramps ΔP , time encodings) and prototyped sequential models (LSTM/GRU) to anticipate power drops.

PROJECTS

RAG-WebSystem – Generative AI system based on Retrieval-Augmented Generation

- End-to-end RAG engine: FAISS-based semantic search over indexed documents, contextual answer generation with an open-source model, exposed via a FastAPI backend and a React + Tailwind interface. GitHub.

Grain Size Analysis of 316L Microstructure – Computer-vision pipeline for grain boundary analysis

- Computer-vision pipeline segmenting grain boundaries in 316L stainless steel micrographs (500 $\times$), comparing thresholding, gradient filters, HED, and a U-Net model, with a Gradio interface. GitHub.

Poverty & Inequality Forecasting in Mauritania – Poverty indicator forecasts

- Forecasts four key poverty indicators from 1996 to 2030 using survey data and interpolation techniques. GitHub.

EDUCATION

Higher Polytechnic School of Nouakchott – ESP - Mauritania

October 2023 – June 2026

State Engineer in Statistics and Data Engineering.

Faculty of Economic Sciences and Management – FSEG - Mauritania

October 2023 – June 2025

Master's in Applied Econometrics and Statistics.

SKILLS

Python • FastAPI • Docker • PostgreSQL • PyTorch • n8n • Data Warehousing • ELT • RAG/LLM

LANGUAGES

Arabic

Native Language

● ● ● ● ●

French

DALF C1

● ● ● ● ●

English

CEFR B1

● ● ● ● ●

VOLUNTEERING

President, ESP Data Club (January 2023 – April 2025)

CERTIFICATIONS

Google Advanced Data Analytics • freeCodeCamp Relational Database • DataCamp Data Engineering